

Approaches to Congestion Control

(End-to-End vs. Network-Assisted Strategies)

1. Overview

Congestion control approaches are primarily distinguished by whether the network layer (IP) provides explicit assistance to the transport layer (TCP/UDP).

2. End-to-End Congestion Control

In this approach, the network layer provides **no explicit support**. End systems must infer the existence of congestion based on observed network behavior.

- **Primary Indicators:**
 - **Packet Loss:** Indicated by timeouts or triple duplicate ACKs.
 - **Delay:** Increased Round-Trip Time (RTT) can signal growing queues.
- **TCP Implementation:** The Internet's default versions of TCP adopt this approach. When loss is detected, TCP dramatically reduces its congestion window.

Inference Principle

The end systems are the "detectors" and "controllers," treating the network as a black box that only provides implicit feedback through arrivals and drops.

3. Network-Assisted Congestion Control

With this approach, routers provide **explicit feedback** regarding the congestion state. This feedback can range from simple binary flags to complex transmission rate mandates.

Forms of Feedback

1. **Simple Bit Indication:** A single bit (e.g., ECN - Explicit Congestion Notification) in the IP header marks a link as congested. Used in architectures like SNA, DECnet, and ATM.
2. **Sophisticated Indicators:** In ATM ABR (Available Bit Rate), routers inform the host of the **maximum transmission rate** they can currently support.

4. Notification Mechanisms

Network-assisted information is typically fed back to the sender in two distinct ways:

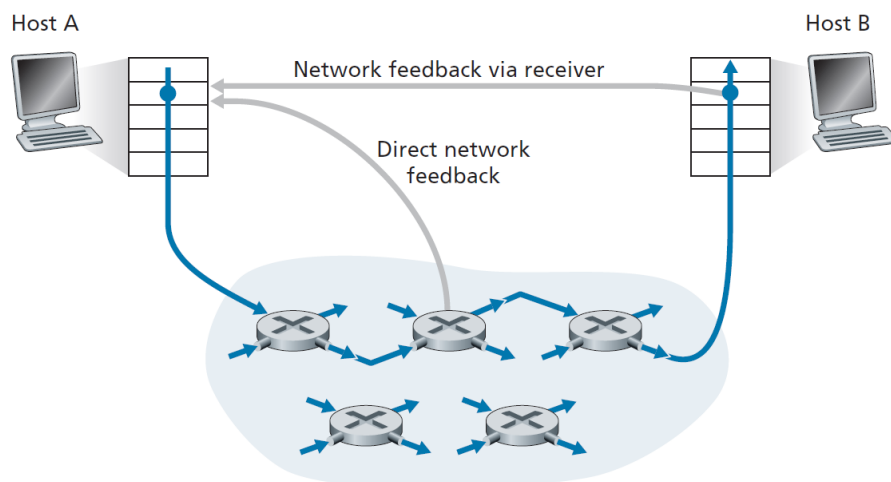


Figure 1: Two Feedback Mechanisms for Network-Assisted Congestion Control

1. **Direct Feedback (Choke Packets):** A router sends a special control packet (a "choke packet") directly back to the sender, saying "I'm congested!".
2. **Router-Marked Packets (Indirect):** A router marks/updates a field in a data packet flowing from sender to receiver. The receiver then notifies the sender upon receipt of the marked packet. This method takes a full **round-trip time (RTT)**.

Full-Duplex Control

In full-duplex systems like TCP, feedback might be carried in reverse-direction ACKs or piggybacked on reverse-direction data segments.

5. Summary Comparison

The Two Philosophies

End-to-End	Network-Assisted
No help from routers. Congestion is inferred (timeouts, etc.).	Explicit help from routers. Congestion is announced (bits, choke packets).
TCP/IP (Default). Simple network core, complex edge.	ATM, SNA, Modern TCP with ECN. Intelligent network core.